SST Topology–Projection–Phase Blind Falsifier v0.2.0

Scientific methods, blind-run assessment, real PKLSA geometry, and dimensionless Euler pilot. 16 October 2026.

Certification status: synthetic calibration and short-time numerical Euler screen passed. NOT a certified physical SST particle model, not a proof of vortex reconnection, and not a native SST Framework v1.0.7 execution.

Executive results

Area	Result	Qualification
Regression suite	16 / 16 PASS	Analytic knot, Hopf linking, winding, synthetic controls, exact Euler manufactured fields
Blind DS01–03	G09 = 3/3 PASS	Previously sealed v0.1 inputs; NOT new replication or revealed truth
Real KnotPlot trefoil N300	Alexander $\|\Delta(-1)\| = 3$	Three generic projections agree; real static source, not dynamic proof
Real PKLSA candidate surface	$\chi = 2$, closed manifold	3906 vertices / 7808 triangles. Genus 0, not a filament
Periodic 3D Euler pilot	Alexander 3 → 3	Same final model time = 0.10, two spatial and two temporal settings
Physical SST certification	UNRESOLVED	Core mapping, SI calibration, independent provider, long-time convergence absent

Central hypotheses and strict meaning

H1: true nonideal 3D reconnection is a change of material vortex connectivity or linkage supported by resolved dynamics and a physically nonideal mechanism. H2: section tangency can create or annihilate 2D cuts even when all 3D loops persist. H3: optical phase zeros can annihilate without corresponding material vortex components. The three synthetic optical maps are byte-identical, hence 2D alone cannot identify H1/H2/H3.

An ideal smooth Euler fluid with no forcing must satisfy material vorticity transport. A topology jump in a synthetic scene or from marker reordering is NOT Euler reconnection. The test intentionally continues through failed/unresolved gates.

Primary literature

Bucher et al., Superluminal correlations in ensembles of optical phase singularities, Nature 651, 920–926 (2026), doi:10.1038/s41586-026-10209-z. The optical singularities are zero-amplitude phase patterns, not material vortex lines.

Mathematical kernels

Alexander determinant: Construct the 2D crossing diagram in three generic projections, identify over/under arcs, build the integer Fox/Wirtiger crossing matrix with coefficients (+2,−1,−1), delete a row and column, and evaluate |det| using exact-integer Bareiss elimination. Expected trefoil = 3; figure-eight = 5; unknotted circle = 1. A determinant equal to 1 does not prove unknot status.

Gauss linking: $Lk(C1,C2) = [1/(4\pi)]\oint\oint [(dr1 \times dr2)\cdot(r1-r2)] / |r1-r2|^3$. For disjoint polygon segments, apply independent 3-point and 5-point tensor Gaussian quadrature; accept only when $|Lk5-\text{round}(Lk5)| < 0.075$, $|Lk5-Lk3| < 0.075$ and the segmentwise centerline gap is positive. Orientation reversal changes the sign.

Ideal Euler: $\nabla\cdot u = 0$, $\omega = \nabla\times u$, $\partial_t\omega + (u\cdot\nabla)\omega - (\omega\cdot\nabla)u = 0$. A material vortex loop must retain its individual circulation; aggregate signed Γ alone is insufficient.

Periodic spectral pilot: $\partial_t u = P(u \times \omega)$, where $P = I - kk^T/|k|^2$ is the Fourier solenoidal projector; the 2/3 dealiasing rule is applied, RK4 steps the field, and material markers are advected by the stage velocities. A Gaussian-smoothed tangent-vorticity tube is an illustrative initial condition rather than a canonically calibrated SST core.

Source data and provenance

Real KnotPlot trefoil centerline: 300 × 3 ordered coordinates; source Drive ID 1iOeQe6-jHcBk_NYNwZe3e-N2DQPwohmI. SHA-256 original a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2. The points have unqualified geometry units, so their transformation into the 2π-periodic box is explicitly dimensionless.

The separate PKLSA `CANDIDATE_SURFACE.npz` has `vertices_m` and `triangles`, defining a triangular mesh. It is closed, manifold, has Euler characteristic 2 and genus zero (assuming orientability). Its surface topology does NOT imply the embedding knot type. Treating its raw vertex order as a vortex centerline would be invalid.

Gate qualifications

Gate	What is actually tested
G09	Knot determinant + linking integral across before/after scene frames
G10	Circulation conservation per stable material filament ID; absent in synthetic scenes
G11	3D Euler divergence and vorticity PDE residual on independently supplied velocity grids
G12	Physical 3D reconnection; cannot PASS for smooth ideal Euler by imposed topology surgery
G13	Spatial / temporal convergence at same time and same core regularization
G14	Independent upstream data and solver replication
G15	Independent physical SI scale and circulation calibration

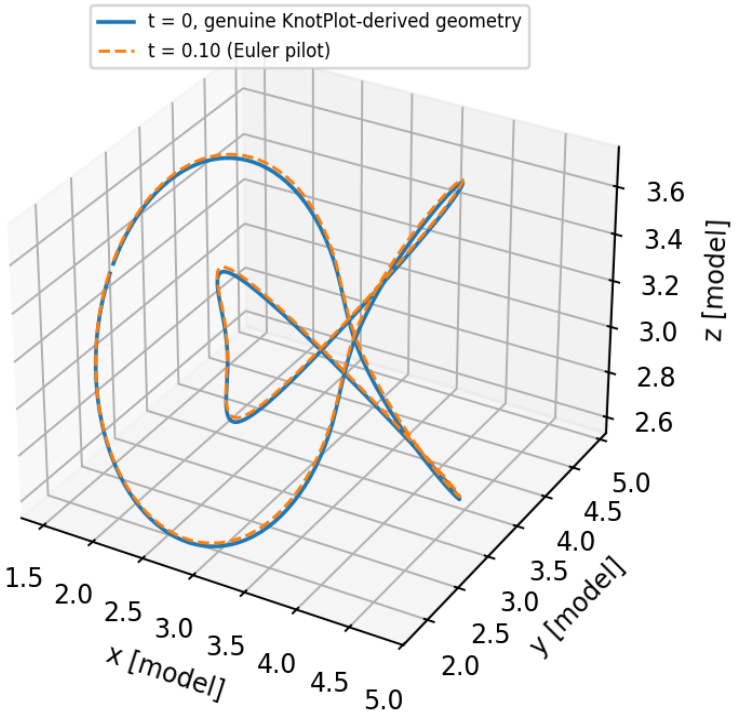
Real-source Euler pilot: three configurations

Source: actual relaxed KnotPlot N300 trefoil. Box: $[0,2\pi)^3$, no forcing, no viscosity, Gaussian width $\sigma = 0.50$ model length units, total model time 0.10, initial dimensionless circulation normalized to unity. These are numerical model units, NOT SI.

Grid / steps	Δt model	ΔE / E0	ΔH / |H0|	Euler FD residual	Spectral div RMS
N=24; steps=4	0.02500	-3.218e-14	7.056e-14	4.279e-05	1.311e-16
N=24; steps=8	0.01250	-1.018e-15	2.302e-15	1.070e-05	1.341e-16
N=32; steps=4	0.02500	-3.361e-14	8.207e-14	4.378e-05	1.760e-16

Short-time topological result in each run: $|\Delta(-1)| = 3$ before and after. At grid 24^3, halving Δt gives an Euler central-difference residual ratio approximately four, as expected for the diagnostic method; this is not evidence of fourth-order RK4 global convergence.

Final-marker RMS difference for N=24 / 4 vs N=24 / 8: 5.3975e-08 model length. At same time / sigma, N=24 / 4 vs N=32 / 4: 0.0001238 model length. Two grids are insufficient for a grid-convergence order.



Plot shows the genuine N300 KnotPlot-derived ordered polygon before and after the N=24 Euler model evolution, in normalized periodic coordinates. The lines almost overlap at this short time.

Blind integrity and physical claims status

The v0.2.0 freeze protocol has SHA-256 853e02e4345e9d261dce93894dd0d6e888311953de7dcb7902dbacec03cbb420. The DS01–DS03 input manifest and AES-GCM encrypted labels are unchanged from v0.1, and are not decrypted by the scientific runner. They remain a useful code-control cohort but not an independent dataset.

Blind predictions: DS01 = section tangency; DS02 = synthetic 3D connectivity change candidate; DS03 = phase-only singularities. Mechanism predictions remain pre-reveal. G09 = PASS in all three because the available sample polygons meet the screen. G10–G15 remain unresolved or diagnostic-only for these sources.

Potential falsifiers of the SST physical interpretation

(F1) A closed, smooth, materially labelled vortex loop displays a verified topology change in a converged ideal Euler integration without loss of regularity: inconsistent with the stated smooth ideal-transport assumptions and signals an implementation/source issue.

(F2) A claimed long-lived trefoil loses its knot signature or becomes numerically unstable as resolution is increased while the regularized physical model and final time are fixed.

(F3) A claimed annihilation seen in a 2D phase image is absent from genuine 3D topology. The event is then a projection or interference artefact, not direct particle annihilation.

(F4) A claimed circulation-conserving reconnection is supported only by aggregate signed Γ while individual tracked material curves change their Γ.

(F5) Different upstream providers or finite-core kernels do not reproduce the proposed stable dynamical state within preregistered tolerances.

Minimum next qualification tranche

1) Register finite-core radius and physical units without calibrating to the answer. 2) Run $24^3/32^3/48^3$ (or finer), two Δt values each, at identical time and σ; verify energy, helicity, divergence, local vorticity transport, filament advection and signed linking. 3) Follow one trefoil through multiple turnover times. 4) Repeat with a second genuinely independent PKLSA provider. 5) Keep reconnection claims CLOSED for smooth inviscid Euler.

References (copy-ready bibitems in accompanying LaTeX)

[1] T. Bucher et al., Nature 651, 920–926 (2026), doi:10.1038/s41586-026-10209-z.
[2] H. K. Moffatt, J. Fluid Mech. 35, 117–129 (1969), doi:10.1017/S0022112069000991.
[3] P. G. Saffman, Vortex Dynamics, Cambridge University Press (1992), doi:10.1017/CBO9780511624063.